

Heart Disease Severity Prediction & Model Comparison

A Modular 11-Step Machine Learning Pipeline with Resampling & Hyperparameter Optimization

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1. Executive Summary & Problem Formulation

Coronary artery disease is a leading global health concern. This project implements a production-style, 11-step machine learning workflow on the multicenter UCI Heart Disease dataset (920 patients). The pipeline encompasses structured data preparation, visual EDA, statistical transformation, feature selection, stratified partitioning, baseline training, SMOTE resampling, cross-validation, hyperparameter tuning, and comprehensive model comparison.

2. Modular 11-Step Pipeline Architecture

The codebase is organized into dedicated, sequential modules matching standard data science practice:

- **Step 1 (Data Preparation):** Handles missing values, cleans categorical strings, and corrects 172 unrecorded zero cholesterol values.
- **Step 2 (EDA):** Generates statistical summaries and distribution charts saved to EDA/.
- **Step 3 (Transformation):** Imputation (median/mode), One-Hot & Binary encoding, and StandardScaler normalization.
- **Step 4 (Feature Selection):** Drops non-predictive identifiers and high-missing columns (ca, thal, slope).
- **Step 5 (Train/Test Split):** Stratified 80/20 train/test split to preserve target class ratios.
- **Step 6 & 7 (Baseline Modeling & Evaluation):** Fits Logistic Regression, Random Forest, and XGBoost classifiers.
- **Step 8 (Data Sampling):** Applies SMOTE to synthesize minority class samples and balance training data.
- **Step 9 (Cross-Validation):** 5-Fold Stratified Cross-Validation across candidate models.
- **Step 10 (Hyperparameter Optimization):** GridSearchCV on XGBoost (learning rate, depth, estimators).
- **Step 11 (Model Comparison & Export):** Benchmarks all models, generates comparative plots, and saves best_model.pkl.

3. Experimental Model Comparison & Benchmark Results

All candidate models were evaluated on the held-out test set (184 unseen samples):

Model Architecture	Sampling Strategy	Test Accuracy	Macro F1	Weighted F1
Logistic Regression	Original Split	60.87%	0.3662	0.5801
Random Forest (Baseline)	Original Split	57.07%	0.2752	0.5077
XGBoost (Baseline)	Original Split	60.33%	0.3855	0.5859
XGBoost (Tuned + SMOTE)	SMOTE Resampled	56.52%	0.4173	0.5787

4. Key Takeaways & Conclusion

- **Impact of SMOTE:** Applying SMOTE improved minority class recall and increased **Macro F1-Score from 0.3855 to 0.4173** on XGBoost, showing improved sensitivity across rare severity stages.
- **Best Model Selection:** Logistic Regression and XGBoost achieved top overall test accuracy (~60.87%), while tuned XGBoost demonstrated superior balanced class discrimination.
- **Delivery Structure:** The repository is equipped with a master Main.py, 11 standalone step scripts, dedicated artifact directories (Datapreparation/, EDA/, TransformedData/, Models/, ModelComparison/), an interactive notebook HeartDiseasePrediction.ipynb, and serialized model artifacts.